ICT Unit B: Computer System Fundamentals

Ch11 The system unit of a Computer System

➢ System unit

- System unit: A case that contains Circuit boards + Power supply + Storage device
- Types of peripherals: Input + Output + Storage + Communication

➢ Component.1 CPU – Central Processing Unit

Difference components

Components of CPU	Sub-components		Functions / Descriptions
Control Unit (CU)			1. Keeps track of the sequence of instruction being processed
			2. Monitor and coordinates all Input/Output operations and system unit
Arithmetic & Logic Unit (ALU)	Arithmetic operation	+ , - , * , /	1. Calculation
	Logical operation	And / Or / Not / Nor	2. Comparison
Registers	**ALU** – General Purpose Register (GPRs)	Accumulator (AX)	1. Data: Load from main memory (into register) + Store from a register (into main memory)
		Base registers (BX)	
		Counter (CX)	
		Data register (DX)	2. Store data temporary in arithmetic operations
	CU – Control Register	Instruction: Instruction Register (IR)	1. Contains the **instructions** to be executed by the CPU
		Instruction: Program Counter (PC)	2. Contains the **memory address** of the **next instruction** to be executed 3. (Value in the PC may be changed as the execution result of the current instruction)
		Data: Memory Address Register (MAR)	4. Hold the **memory address** of which data is to be transferred to or from
		Data: Memory Data Register (MDR)	5. Contains the **data** to be written into or read from the location specified in MAR
	Status Register (SR)	Indicating the status after the execution of instruction (conditional testing and program branching) ER = Evaluated Result	1. Zero flag 1:ER is zero 0:ER is 1
			2. Sign flag 1:ER is < 0 0:ER is > 0
			3. Parity flag 1:ER is an **odd** parity
			4. Carry flag 1:ER contains a carry 0:does not
			5. Overflow flag 1:ER has overflowed (Too large to be stored in a memory unit)

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- CPU Speed (Clock rate = Clock cycle pers second) (e.g. 2.00 GHz)
- Word width: Number of bits of data + Instruction that CPU can **handle** per unit time (e.g. 64-bit)
- Bus width: Number of bits of data can be **transferred** at a time (e.g. 64-bit data bus)

Extra:

1. **Pipelining Technology (Multi-core)**: Core that can work in parallel efficiently
   → Reduce speed-gap between RAM and CPU
2. **Paralleling processing**: Multi-cores work on the same time

➢ Component.2 Memory – RAM vs ROM vs Cache Memory vs CMOS

⇒ Store data + instruction → CPU executes

1. **RAM – Random Access Memory (Volatile)**

   ⇒ **Main memory** (Memory that can be manipulated directly by CPU) in a computer

   ⇒ Temporary hold the **data and instruction** of application software and operating system (OS)

	Dynamic RAM (DRAM)	Static RAM (SRAM)
Clock rate	Lower	Higher
Cost	Lower	Higher
Power consumption	Higher	Lower
Storage capacity (per every physical size)	Smaller	Larger
Application	Main memory, Game console	CPU cache, Buffers, PDA

2. **ROM – Read-Only Memory (Non volatile)**

   ⇒ Store **permanent information** supplied by the manufacturer

   ⇒ **Bootstrap program**: a sequence of instructions that **load and initialize OS**

   1. EPROM Erasable Programmable Read-Only Memory
   2. EEPROM (Flash Memory) Electrically Erasable Programmable Read-Only Memory

3. **Cache Memory (High-Speed memory) (Small quantity) (Integrated in CPU)**

   ⇒ Stores data and instructions that recently used by CPU

   ⇒ **CPU requests in instruction or data**: Search the cache memory first (✕ main memory) → Required data found → CPU acquire the data → Proceeds to the next task

4. **CMOS memory (Complementary Mental Oxide Semiconductor memory)**

   ⇒ Store data for a long time (Even PS switched off)

   ⇒ Information of the computer system (e.g. HDD, Memory capacity, Display configuration)

➢ Execution.1 Machine Cycle – Execute an instruction in a CPU

1. **Fetch** Read the next instruction: Main memory → Instruction Register (IR in CU)
2. **Decode** Identify the **operation code + operands** in an instruction
3. **Execute** Interpret the **operation code** and **perform the required operation**

Program instruction execution	**Interrupt sub-cycle:**
1. *CU* issues a command to load the next instruction 2. *CU* sends a 'read' control signal to the main memory via control bus 3. *PC* copies the address of the required instruction to *MAR* 4. *MAR* passes the address to the main memory via the address bus 5. The main memory returns the addressed instruction to *MDR* via the data bus 6. The instruction in *MDR* is then loaded into *IR* for execution	1. CPU saves the current process status in the stack (Stack memory) 2. Handles the interrupt immediately

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Ch12 Input and Output device

➢ Input device

⇒ Enter data into computer → Converts it into computer readable form

Input	Device	Explanation
Command	**Pointing device**: Control cursor / pointer → Trigger actions	
Text	**Keyboard**: Enter data / commands	
Image	1. **Scanner**: Project light onto the object to be scanned, captures the reflected light and converts the signals into bitmap image 2. **Optical readers** (Bar code readers, optical mark readers): Read predetermined codes → translate them into digital data 3. **Bar code reader** (Mini-scanner): Captures images of bar codes and translate them into digital data 4. **Optical mark reader**: Use with OMR software 5. **Digital camera**: Uses light-sensitive processor chip → Capture scenes → Convert to digital images	1. **Quality (scanned image)** ∝ (Resolution + colour depth + Accuracy) 2. **Resolution (scanned image)**: dots per inch (dpi) 3. **Optical character recognition (OCR) software**: Read scanned image of text document → Translate it into text file 4. **Optical Mark Recognition (OMR) software**: Recognizes specific handwritten marks 5. **Number of light-sensitive photosites** ∝ Resolution of digital image captured
Audio	**Microphone**: Record sound	
Video	**Digital video camera**: Uses light-sensitive processor chip → Capture the moving images and Sequence of high-quality digital images	
Others	Mouse, trackball, touchpad, track point, joystick, touch screen, digitizing tablet, handwriting board **Biometric devices**: Convert characteristics of a person → Digital Code (e.g. Fingerprint scanner, Speech Recognition system, Iris recognition system)	**Mouse**: Mechanical mouse, Optical mouse

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➢ Extra: Application

- CCD: **Charge Coupled Device** (Convert analogue video signal ⇔ Digital video signal)

Devices	Usages
Bar code reader **Thermal printer**	Point-of-sale terminal
Cheque deposit machine (ATM card reader)	Scans the cheques one by one and prints receipt slip using thermal printer

- Dot pitch vs dpi → Determine the **quality**
- Dot pitch: Monitor Dpi: Scanner, Ppi: Printer

➢ Output device

⇒ Translates computer process + Information → Narrative form

<table>
<tr><th>Output</th><th>Devices</th><th>Explanation</th></tr>
<tr><td>Image</td><td>Visual display unit (VDU) (Monitor): Shows computer output instantly on screen (CRT monitors and LDC monitors)<table><tr><td>Advantage of LCD</td><td>Advantage of CRT</td></tr><tr><td>• Slimmer
• Clearer display
• Power consumption↓</td><td>• Contrast ratio↑</td></tr></table></td><td>1. CRT: Cathode ray tube
2. LCD: Liquid crystal display
3. Quality (Monitor's display) ∝ (Monitor's resolution + Dot pitch + Contrast ratio)
4. Contrast ratio: Measurement of the difference in light intensity between the brightest white and the darkest black
5. Refresh rate: Number of times the whole screen is redrawn in one second
6. Pixel: Smallest display unit of the monitor screen</td></tr>
<tr><td>Sound waves</td><td colspan="2">Speakers: Convert audio signals of computer → Sound</td></tr>
<tr><td>Hard copies</td><td>Printer: Produces text and graphics on physical medium
Multifunction printer: Communication and printing devices (e.g. printer, telephone)</td><td>Performance of printer: Measured by the Printing resolution + Printing speed</td></tr>
<tr><td></td><td colspan="2"><table><tr><th>Type of printer</th><th>Mechanism</th><th>Special feature</th></tr><tr><td>Dot-martix printer</td><td>Strike pins on an ink ribbon to form dotted image on the paper</td><td>• Printing cost↓
• Able to produce duplicates simultaneously</td></tr><tr><td>Inkjet printer</td><td>Form images by spraying tiny ink drops onto paper</td><td>• Quality colour printing↑
• Printing cost↑</td></tr><tr><td>Laser printer (Fast)</td><td>Use laser beam to produce an electrostatic film on drum The toner stuck on the charged areas then transferred and fused onto paper</td><td>• Printing quality↑
• Printing cost < that of inkjet printers</td></tr><tr><td>Thermal printer (Cheap)</td><td>Produce printed image by selectively heating thermal paper when the paper passes over the thermal print end</td><td>• Commonly used in point-of-sale</td></tr><tr><td>Plotter (Quality)</td><td>Similar to inkjet printer</td><td>• Printing cost↑</td></tr></table></td></tr>
<tr><td>Video (Projector)</td><td>Video projector: Projects the computer display onto large screen (LCD projectors and DLP projector)</td><td>DLP: Digital Light Processing</td></tr>
</table>

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Ch13 Secondary Storage Devices

➢ Secondary storage device

⇒ Store Data + Software on a computer system

- **Access time**: Time taken for the device → **Search + Read** required data on storage medium
- **Seek time**: Time taken for **Access Arm** → **Move the read/write head** over the right track + sector
- **Drive rotational speed (rpm)**: Number of revolutions (By disk platters) per minute
- **Data transfer rate (bps)**: Amount of data transferred (Main memory ⇔ Storage device) per second
- **Storage capacity**: Maximum amount of data that can be stored on a storage medium

Types	Example
1. Magnetic Storage Device	HDD (Magnet coating + Aluminium glass + Ceramic
2. Optical Storage Device	DVD, Blu-ray Disc
3. Other Types	Memory cards (Flash memory), USB flash drives

1. Magnetic Storage Device

Devices	Description
Hard Disk (~ 2TB)	• Fast + Durable + High-capacity storage medium • With platters with magnetic coating • Performance affected by : Seek time + Drive rotational speed
Magnetic tape (~ 5TB)	• Sequential access device → Data backup purpose
Floppy Disk (~ 1.44MB)	• Electromagnetic charge on surface → Backup and transfer a small amount of data between computers

2. Optical Storage Device

Devices	Description
Compact disc (CD-ROM, CD-R, CR-RW)(~ 750MB)	Plastic disks coated with a metallic surface (CD: 1x = 150KiBps) (DVD: 1x = 1.32MiBps) (Data representation: **Microscopic Pits**)
Digital versatile disc (DVD, HD-DVD)(~ 17GB)	
High-definition digital versatile disc (HD-DVD) **Blue-ray Disc (~ 50GB)**	

3. Other forms / Mixture

Devices	Description
Magneto-optical (MO) disk (~ 9.1GB)	Durable + Removable + Rewritable
Memory card (~ 8GB)	• Small + Light + Rewritable Portable + Quiet • Hold data in an electronic format • Low power consumption + Direct access (e.g. Compact Flash (CF) card)
USB flash drive (~ 16GB)	Internal memory card + USB interface adapter (✓ Security control)
Microdrive (~ 8GB)	E.g. Compact Flash II
Hybrid hard-disk (~ 250GB)	HDD + SDD (Solid State Drive)

4. Network Storage

- Files attached on a file server on the web, with magnetic tape/ HDD as storage system
- **NAS**: Network-Attached Storage, **SAN**: Storage Area Network

★ File size (About software): 1MB = 1024B ; Data transmission speed (About hardware): 1MB = 1000B

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Ch14 System Software and Application Software

➢ Relationships between hardware, software and uses

- **Computer software**: Sequence of instructions → Instructs a computer → Perform specific tasks

Application software	→ Perform a certain task
System software	Coordinates the application software and hardware → ✓ Application software to function efficiently

- **User Interface**: Means for software and user interaction

Type.1 System software

Type of system software	Function
1. Operating system	Manage system resources → Run application software
2. Utility program	Perform specific tasks (e.g. System security, Data protection)
3. Driver program	E.g. ATi Catalyst Control Centre

- Bootstrap program (BIOS – Basic Input/Output System) (Stored in the ROM)

  ⇒ Tests hardware in computer system + Loads OS into main memory (Start-up process)

➢ System Software – Operating System (OS)

Type	Function	Example
1. Single-user	1 User per time	Disk OS (DOS)
2. Single-tasking	1 Program per time	MAC II
3. Multi-user	> 1 User per time (Concurrent)	Microsoft Window 8
4. Multi-tasking	> 1 Program per time	Mac OS X Lion 10.7
5. Multi-processing	> 1 CPU per 1 program	Dual Core Gaming for GTA IV

Categories of OS:

Category	Feature	Example
1. Network OS	Run on **Network servers** → Provide network services for the workstation	UNIX, Linux, Window Server 2008
2. Desktop OS	Run on **standalone** 獨立的 **Desktop / Notebook computer**	DOS, Mac OS X, Windows 8
3. Mobile OS	Run on **Portable computing devices / Wireless communication device**	Palm OS, iOS, Android

Functions of OS:

Functions	Description	Example
1. Interface platform	✓ Run application software (By computer)	Java Platform
2. File management	1. Transfers files (Main Memory ⇔ Secondary Storage) 2. Manages file folders 3. Allocates the secondary storage space 4. Provides file Protection + Recovery	Mac Finder
3. Memory management	Allocates the use of RAM to requesting processes	Control panel
4. Device configuration	Controls peripheral devices connected to the computer	
5. Network communication management	Sets up and coordinates the network communication activities	

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➢ System Software – Utility program

⇒ **Protect** data files + **Refine** computer system + **Manage** computer resources → Enhance an OS

Types of Utility program:

Utility program	Function	Example
1. Anti-virus software	Detect + Remove viruses (On computer systems)	NOD32
2. File manager	1. Organize files / directories systematically 2. Facilitate efficient file access	Nautilus
3. Data recovery utility	Recover data that is corrupted / accidentally deleted	elMAGE
4. File compression utility	Compress files by a compression algorithm → Produce a compressed file in a smaller file size	WinZip
5. Backup utility	Make a duplicate of Selected files/ Entire hard disk	Norton Ghost
6. Program uninstaller	Remove Application + All associated system files	Revo Uninstaller
7. Disk scanning program	Detect + Correct logical errors (On storage devices)	Fixit Utility
8. Disk defragmenter	1. Reorganize scattered data (On a hard disk) 2. Restore each file in a contiguous location	Diskeeper
9. System monitoring software (Tune & monitor)	1. System performance 2. Resources activities of CPU, disks, memory	Tune-up utility

➢ System Software – Driver program (Device driver)

• Small program → A specific peripheral device → Communicate with a computer system

Type.2 Application software

• Cross-platform: Development of 1 program for > 1 operating system (e.g Java Virtual Machine)

Types	Function	Example
1. Productivity software	Perform common tasks in schools & offices	Office Word 2010
2. Communication software	Communicate + Share information with others	Windows Live Messager
3. Multimedia software	Create + Mange multimedia files	iTunes
4. Entertainment software	Play computer games + Run multimedia files	XBMC (~ Media Centre)

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Ch15 Computer Systems

Concept.1 Classifications of computer

	Desktop computer	Notebook computer	Handheld computer
Processing capability	Low ~ very high	Low ~ average	Very low
Ability of extension	✓	✕	✕✕
Cost	Low ~ very high	Average ~ high	Low ~
Portability	✕	✓	✓✓
Hours of operation	Weeks	Few hours	
Durability	High	Average	
Power consumption	Average ~ high	Low	Very Low
Application	Home use / Office use	People on the move	Business people

Keywords: Expansion cards (Slots), Ports (e.g. VGA ports, USB ports)

Types	Description	Examples
1. Personal Computer → Handle data IPO (Input-process-output)+ Storage	Desktop computer	Single-user computer system
	Workstation	Powerful desktop computer for specialists
	Notebook computer (Laptop computer)	Small and light-weight personal computer for mobility (✕ ergonomic ∵ Positions of monitor & keyboard: ✕ adjusted separately)
	Handheld computer	Pocket-sized device
	Network computer	Low-cost computer → Business network applications
2. Mainframe computers (Huge + Powerful)	Runs with tailored OS → Control over a computer network	Manage large-scale Database system + Server-side applications
3. Supercomputers (Most powerful)		Handle extremely complicated calculation (e.g. Weather forecast + Medical research)

Concept.2 Characteristics of computer system

Types	Description	Examples / Advantages / Targets
1. Single-user system	Process a user's requests at a time	E.g. Personal computers and handheld computing devices
2. Multi-user system	✓ > 2 users operate simultaneously (Indeed, 1 by 1: Time sharing technology)	E.g. Network servers and mainframe computers
3. Batch processing system	Executes batch jobs at a scheduled time **automatically** (✕ Human interaction)	✓ Utilization of computer resources ↑ ✓ Idling time of the computer ↓ ✓ Schedule time-consuming jobs to non-peak hours
4. Real-time system	Processes data **immediately** after the data is entered Monitor command input every moment	
5. Online interactive system	Processes data (Include update data) **after a designate time frame** after the **data is entered**	
6. Parallel computer	Consists of **a number of CPUs** + **a memory system**: Operate independently + in parallel in different modules	→ Run multiple tasks **simultaneously**/ Run a **single job consisting of many parallel processes**/ Run a process with **degree of pipelining**
7. Distributed processing system	Collection of computers interconnected by a high-speed network: Contains sophisticated software (**Load manager**) → Detects + Instructs cooperation of computers → Accomplish a task ✓ Scalable + Fault-tolerable	

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